Design and Simulation of a Hybrid Wired-Wireless Network Using Cisco Packet Tracer

Table of Contents

1. Introduction....4

2. Objectives and Requirements....4

3. Deriving the Block IP Address....5

4. Subnet Planning and Allocation (VLSM)....5

5. Network Topology Design in Cisco Packet Tracer....6

6. Router and Sub-Interface Configuration....9

7. Routing Implementation Using RIP v2....11

8. Wireless Configuration and WPA2 Security....13

9. DHCP Configuration for Wireless Clients....15

10. Verification and Connectivity Testing....16

11. Evaluation of Design and Practical Learning....19

12. Conclusion....20

References....21

Appendix:....24

List of Figures

Figure 1 Network Topology Design in Cisco Packet Tracer 6
Figure 2 Subnet A (Wireless) 7
Figure 3 Subnet B wired connections 8
Figure 4 Subnet C (Wired) 8
Figure 5 Router-Router Connections 9
Figure 6 R0: Router A: Configuring serial interfaces (Router-Router link) 10
Figure 7 Router B: Configuring serial interfaces and interface connected to subnet B 10
Figure 8 Router C: Configuring serial interfaces and interface connected to subnet C 11
Figure 9 Router A: Routing Implementation Using RIP v2 12
Figure 10 Router B: Routing Implementation Using RIP v2 12
Figure 11 Router C: Routing Implementation Using RIP v2 13
Figure 12 Wireless Configuration and WPA2 Security: Access Point1 13
Figure 13 Wireless Configuration : LAptop_A1 14
Figure 14 Laptop_A1: Connecting to Accesspoint A1 14
Figure 15 Laptops Successfully conncted to Accesspoint 14
Figure 16 R0: Configuration DHCP pool for wireless IP configuration 15
Figure 17 Successfully assigned Ips to wireless devices in Subnet A using DHCP 16
Figure 18 Ping from Laptop_A1 to its default gateway 17
Figure 19 Ping from Laptop_A1 to server B 17
Figure 20 Ping from Laptop_A1 to PC_C1 in subnet C 18
Figure 21 Simulation: PC_C2 in subnet C to PC_B3 in subnet B 18
Figure 22 Successful: PC_C2 to PC_B3 19

List of Tables

Table 1 Overview of Network Topology 4
Table 2 Subnet Layout 6

1. Introduction

Digital communication within contemporary settings depends heavily on computer networks where the various devices can be interconnected to share data and make use of common resources. The networks are increasingly integrating the wires and wireless technologies as the organisational needs change to maintain the balance among stability, speed as well as mobility. Wired networks are still needed to ensure the stability of throughput in servers and stationary computers, and wireless networks are needed to provide flexibility and ability to utilize the recent trend of mobile computing (Raveendran and Tabet Aoul, 2022).

The coursework has a direct relation to these changing needs because it involves a practical design and implementation of a hybrid local area network with the use of Cisco Packet Tracer. The work entails developing a topology that has several subnets, IP addressing through a classless method, routing protocols, secure wireless connection, and end-to-end communication testing. This is to be achieved with the purpose of converting theoretical knowledge into practical ability by showing proper addressing, segmentation, routing and security in a working simulation (Qi et al., 2025).

2. Objectives and Requirements

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Network Component	Device Count	Purpose
Routers	3 (Router1, Router2, Router3)	Routing between Subnet A (Wireless), Subnet B (Wired), Subnet C (Wired), and inter-router networks
Switches	2 (Switch B, Switch C)	Providing wired access for Subnet B and Subnet C
Wireless Access Point	1 (AP_A)	Providing wireless connectivity for laptops in Subnet A
Wired PCs	7	End-user hosts in Subnet B and Subnet C
Servers	2	Network services in Subnet B and Subnet C
Wireless Laptops	3	Wireless end-user devices in Subnet A

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Subnet	Mask	Address Range	Number of Hosts	Gateway
Subnet A – Wireless	/28 → 255.255.255.240	191.91.56.1 – 191.91.56.14	4+ hosts	191.91.56.1
Subnet B – Wired	/28 → 255.255.255.240	191.91.56.17 – 191.91.56.30	4 hosts	191.91.56.17
Subnet C – Wired	/27 → 255.255.255.224	191.91.56.33 – 191.91.56.62	5 hosts	191.91.56.33
Subnet #30 – Router Links	/30 → 255.255.255.252	191.91.56.64 – 191.91.56.75	2 hosts per link	N/A

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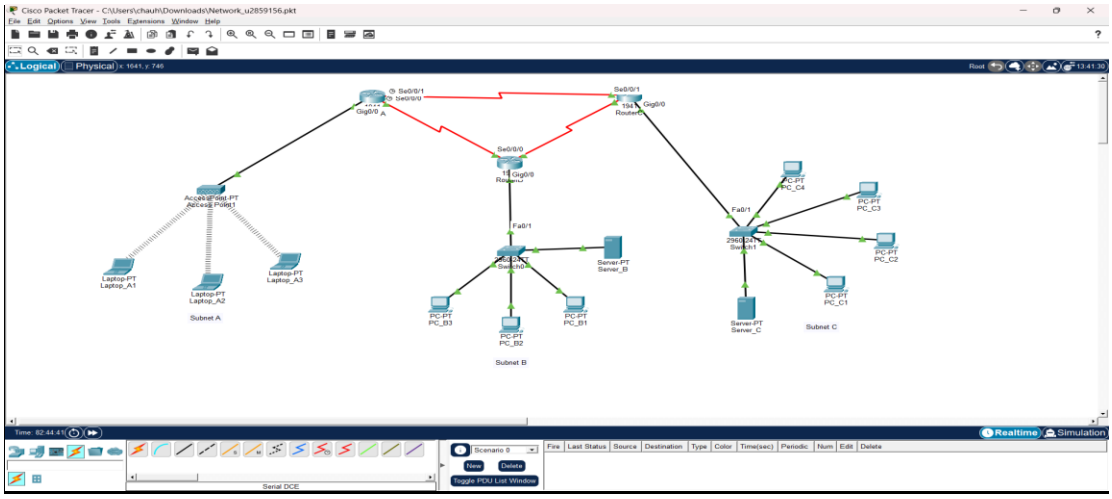

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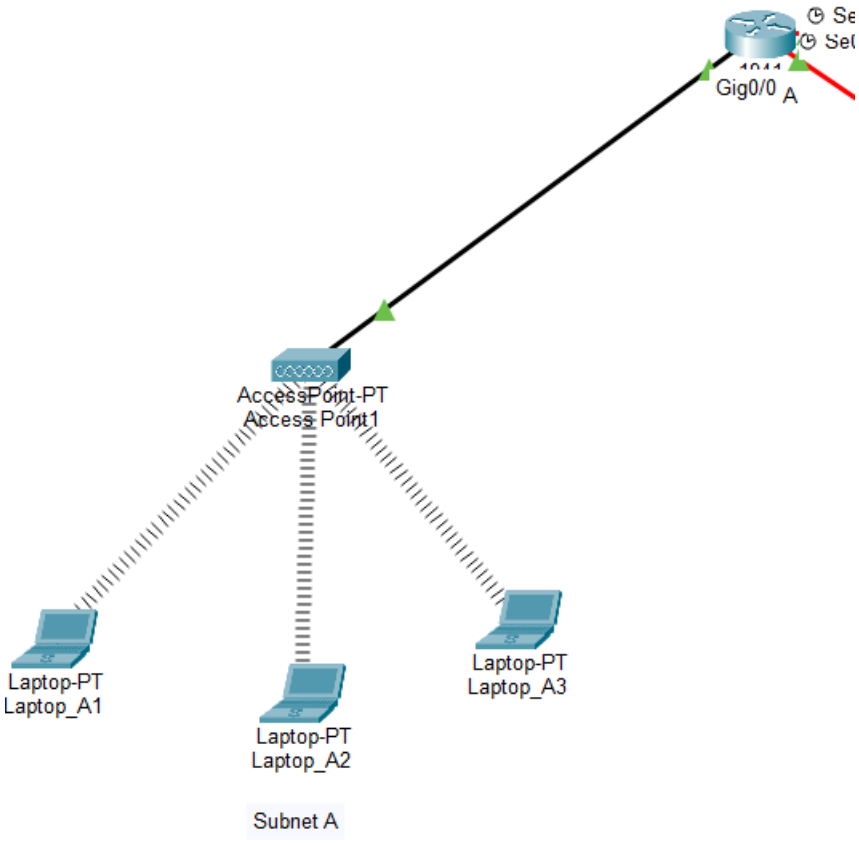

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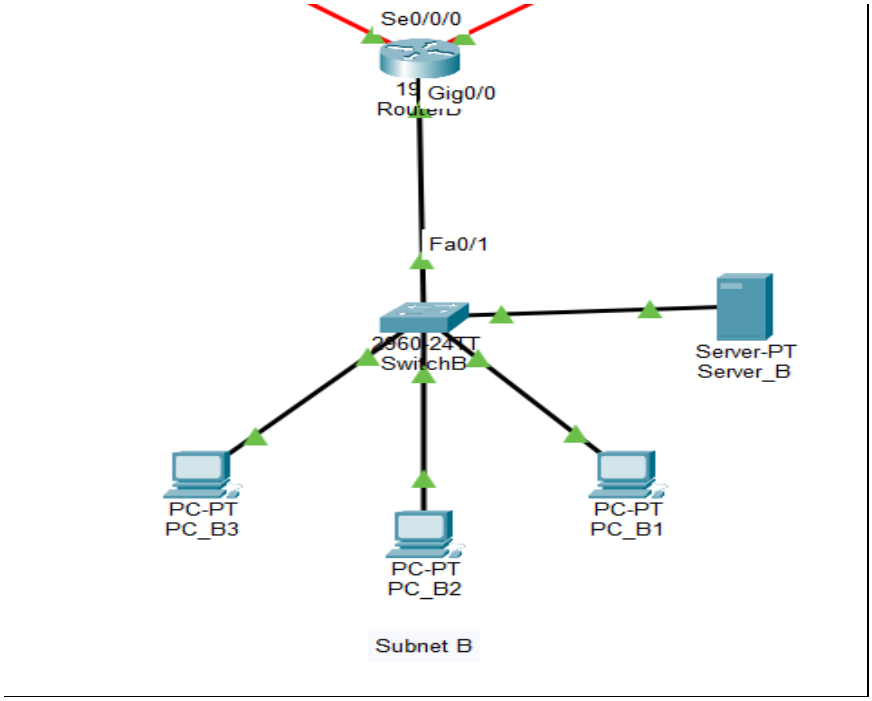

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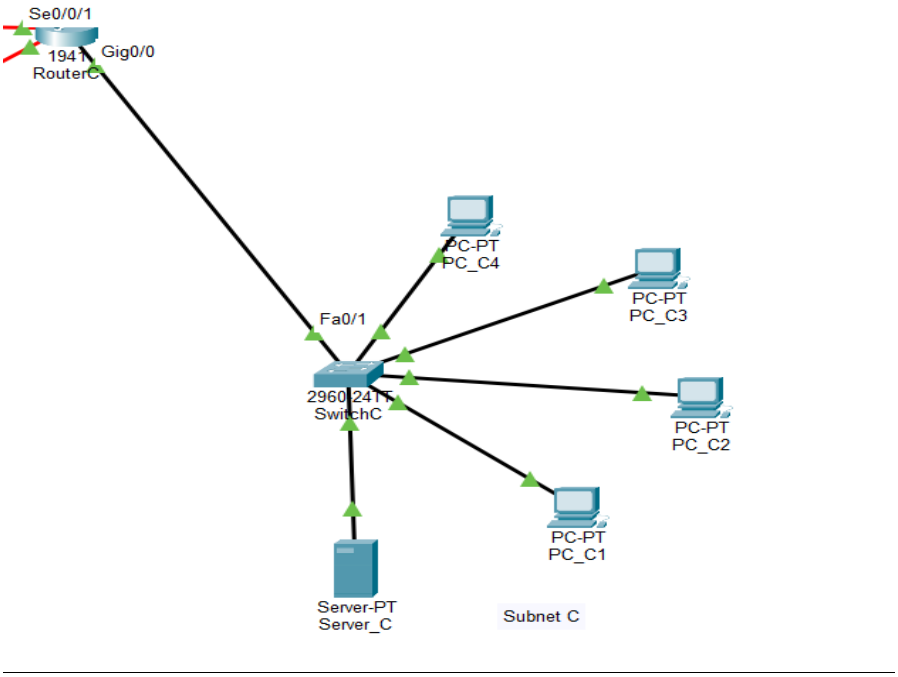

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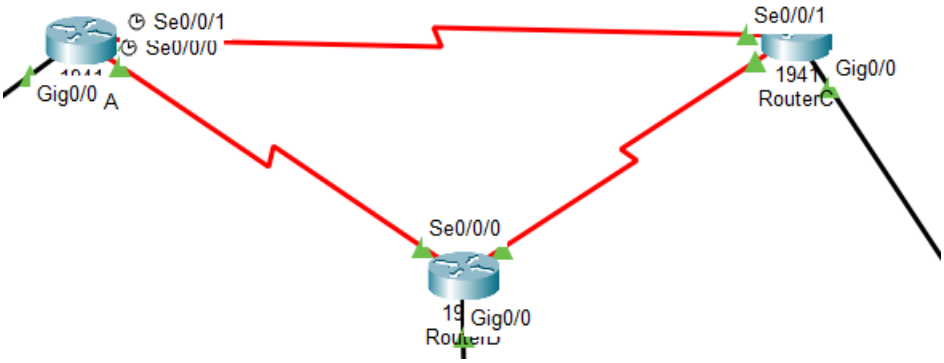

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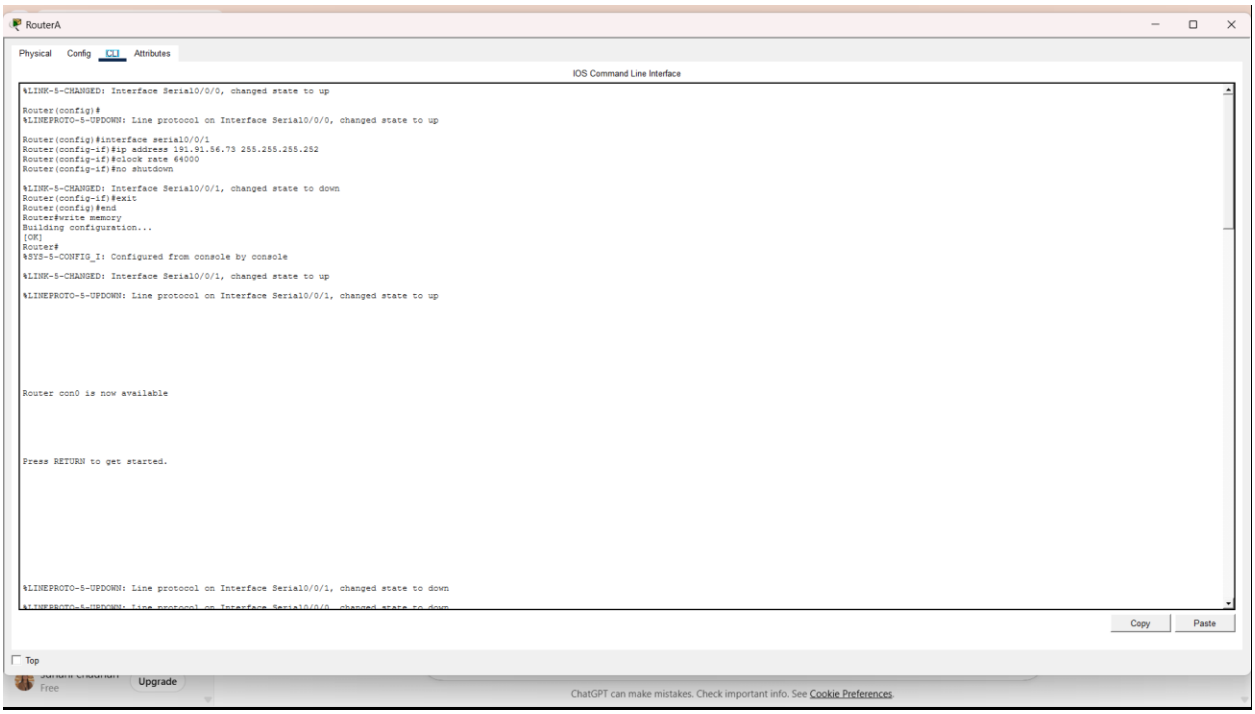

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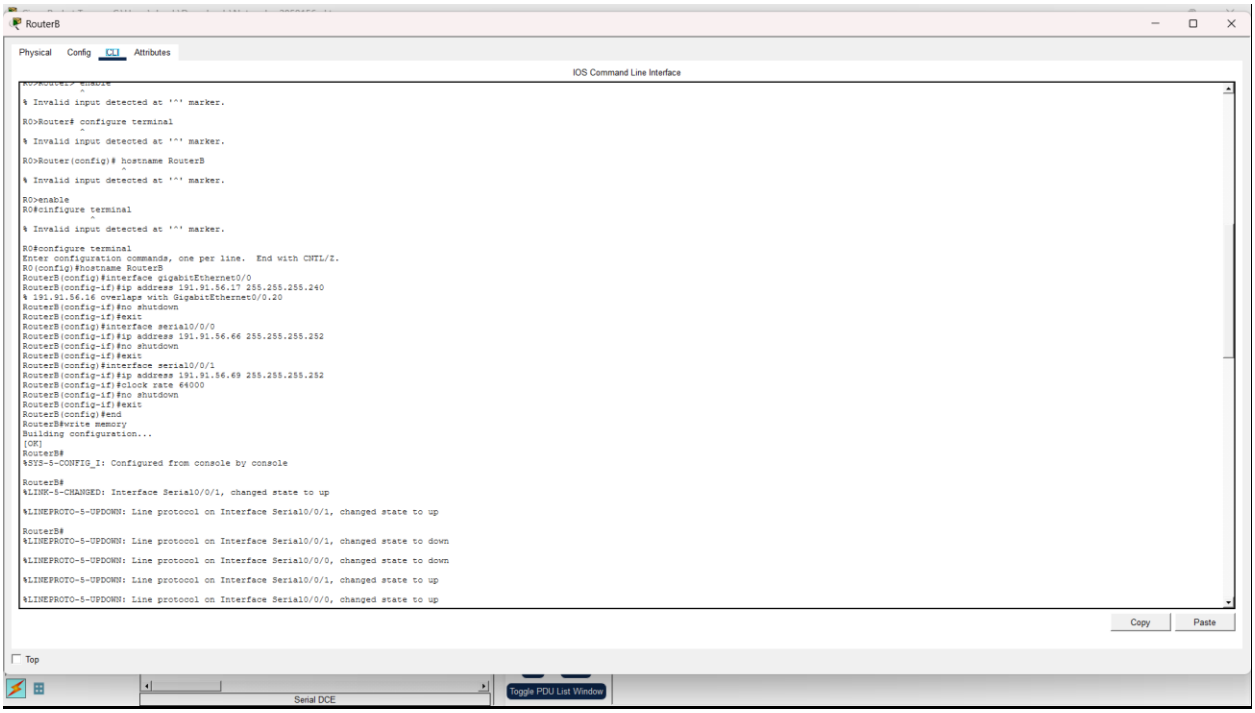

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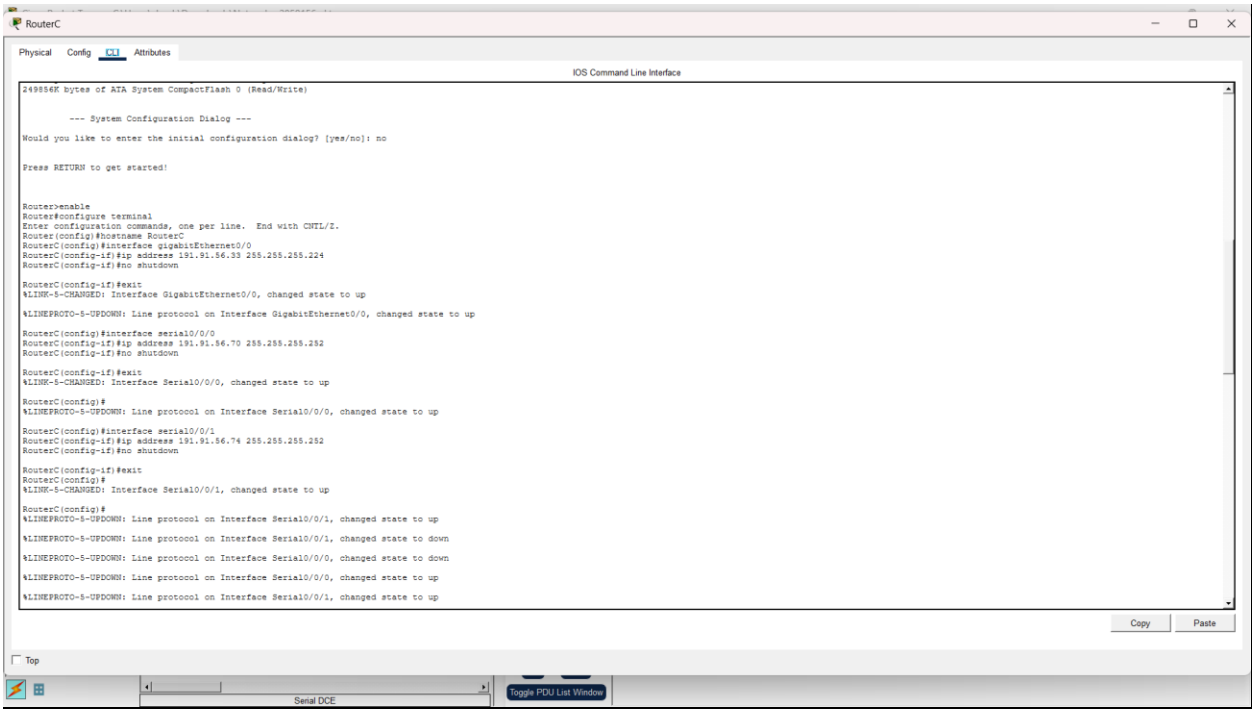

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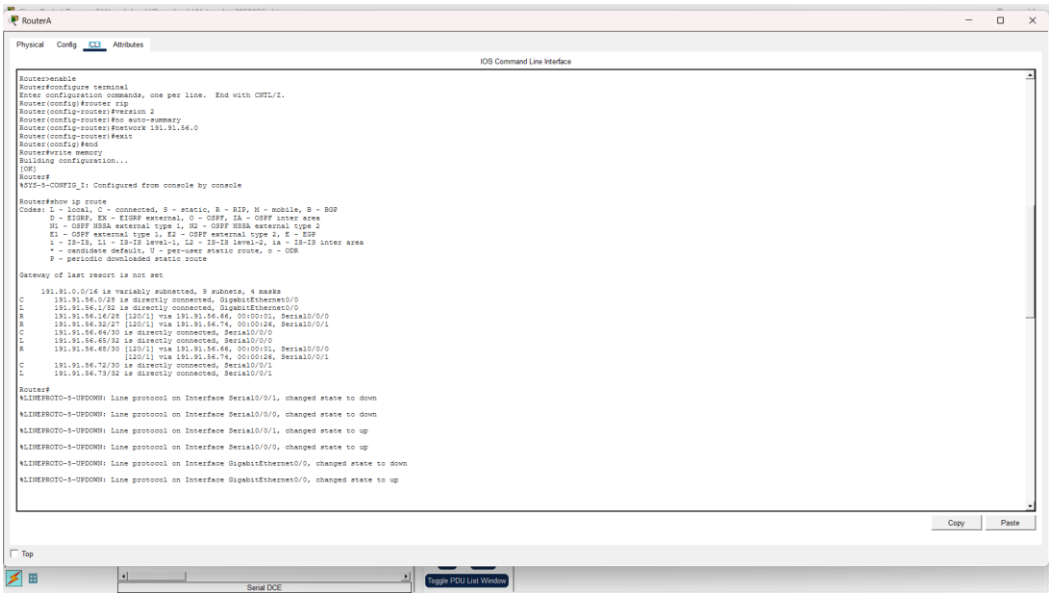


```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#no auto-summary
Router(config-router)#network 191.91.56.0
Router(config-router)#exit
Router(config)#end
Router#write memory
Building configuration...
[OK]
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     191.91.0.0/16 is variably subnetted, 9 subnets, 4 masks
C       191.91.56.0/28 is directly connected, GigabitEthernet0/0
L       191.91.56.1/32 is directly connected, GigabitEthernet0/0
R       191.91.56.16/28 [120/1] via 191.91.56.66, 00:00:01, Serial0/0/0
R       191.91.56.32/27 [120/1] via 191.91.56.74, 00:00:26, Serial0/0/1
C       191.91.56.64/30 is directly connected, Serial0/0/0
L       191.91.56.65/32 is directly connected, Serial0/0/0
R       191.91.56.68/30 [120/1] via 191.91.56.66, 00:00:01, Serial0/0/0
                        [120/1] via 191.91.56.74, 00:00:26, Serial0/0/1
C       191.91.56.72/30 is directly connected, Serial0/0/1
L       191.91.56.73/32 is directly connected, Serial0/0/1

Router#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
```

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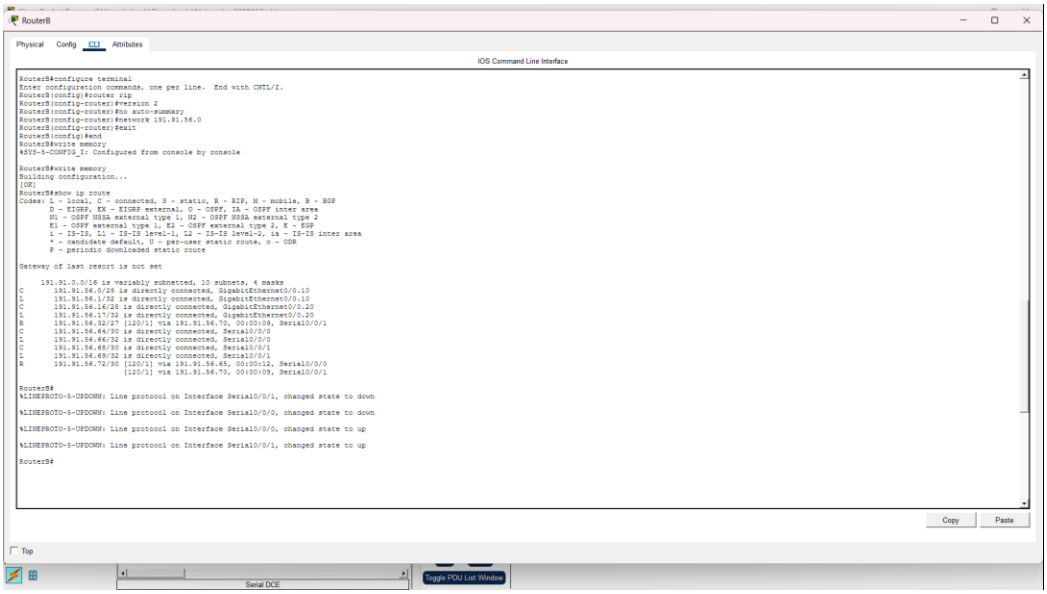


```
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
RouterB(config)#router rip
RouterB(config-router)#version 2
RouterB(config-router)#no auto-summary
RouterB(config-router)#network 191.91.56.0
RouterB(config-router)#exit
RouterB(config)#end
Router#write memory
%SYS-5-CONFIG_I: Configured from console by console

Router#write memory
Building configuration...
[OK]
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     191.91.0.0/16 is variably subnetted, 10 subnets, 4 masks
C       191.91.56.0/28 is directly connected, GigabitEthernet0/0.10
L       191.91.56.1/32 is directly connected, GigabitEthernet0/0.10
C       191.91.56.16/28 is directly connected, GigabitEthernet0/0.20
L       191.91.56.17/32 is directly connected, GigabitEthernet0/0.20
R       191.91.56.32/27 [120/1] via 191.91.56.70, 00:00:09, Serial0/0/1
C       191.91.56.64/30 is directly connected, Serial0/0/0
L       191.91.56.66/32 is directly connected, Serial0/0/0
C       191.91.56.68/30 is directly connected, Serial0/0/1
L       191.91.56.69/32 is directly connected, Serial0/0/1
R       191.91.56.72/30 [120/1] via 191.91.56.65, 00:00:12, Serial0/0/0
                        [120/1] via 191.91.56.70, 00:00:09, Serial0/0/1

RouterB#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/1, changed state to up

RouterB#
```

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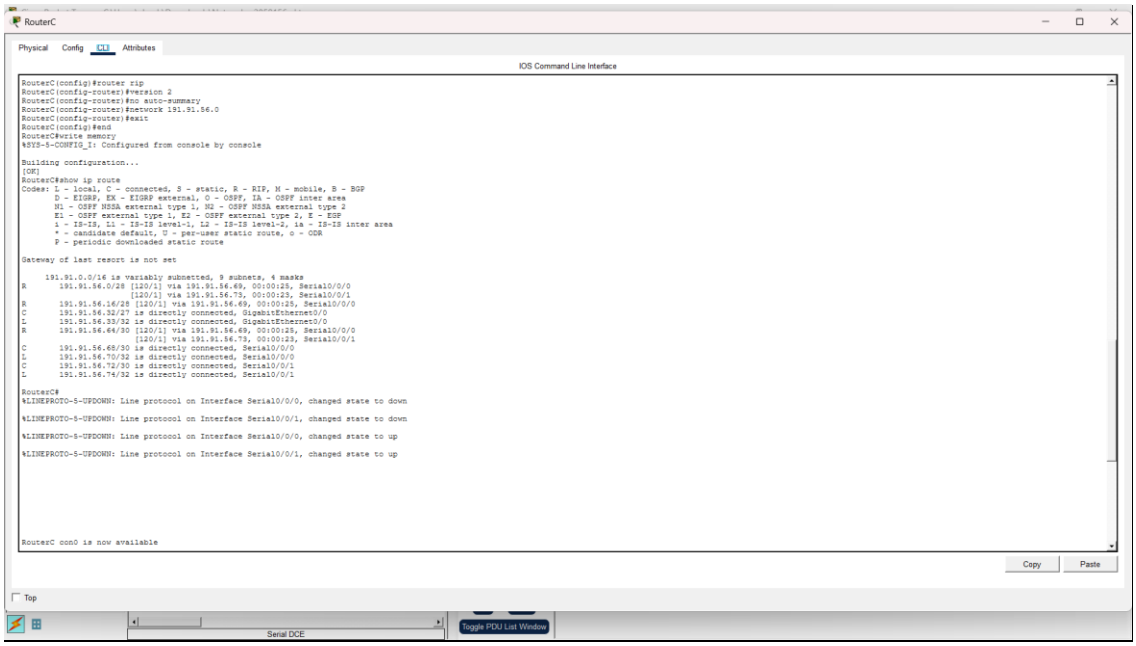

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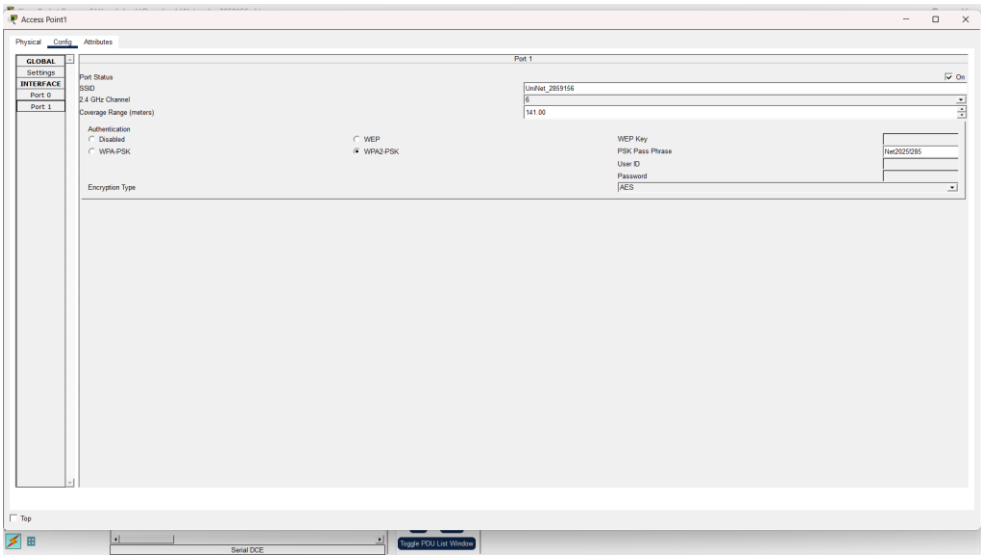

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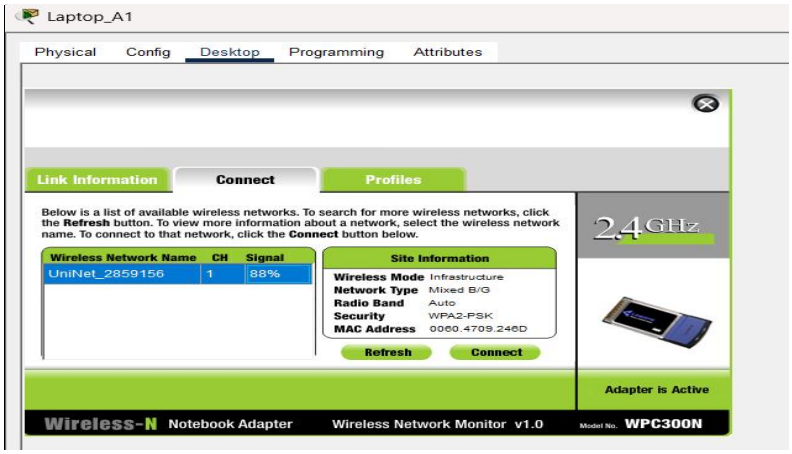

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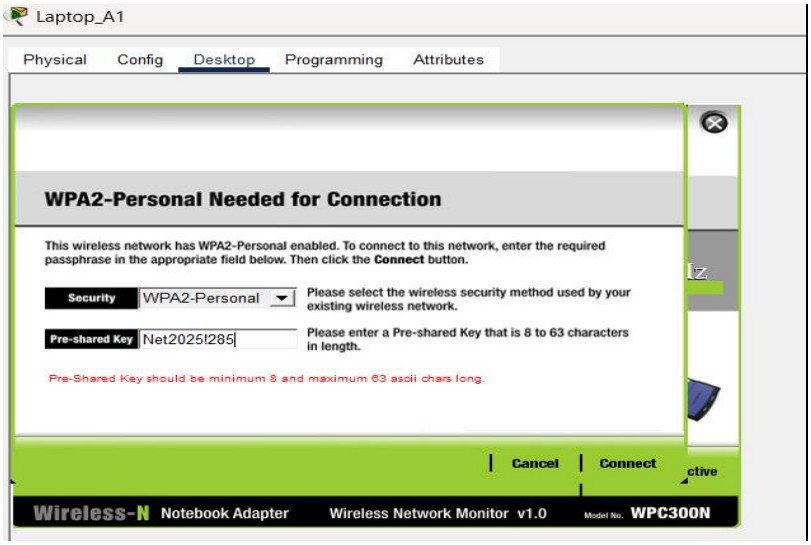

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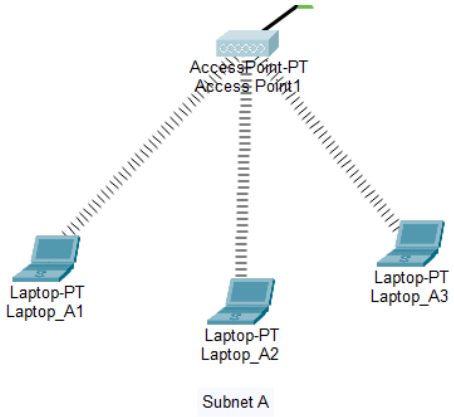

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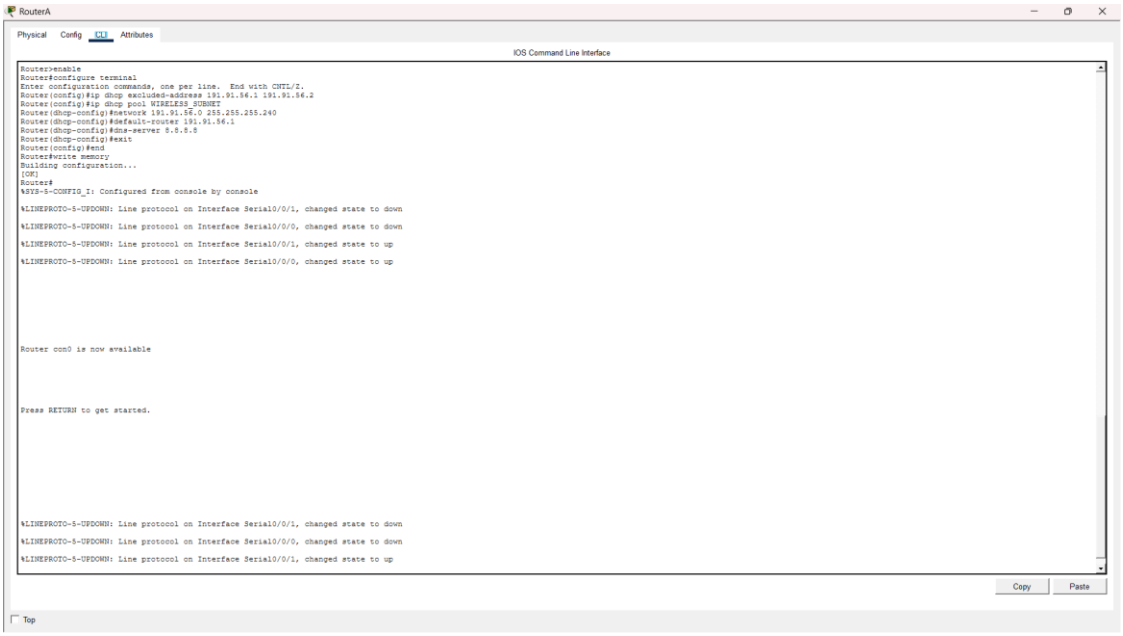

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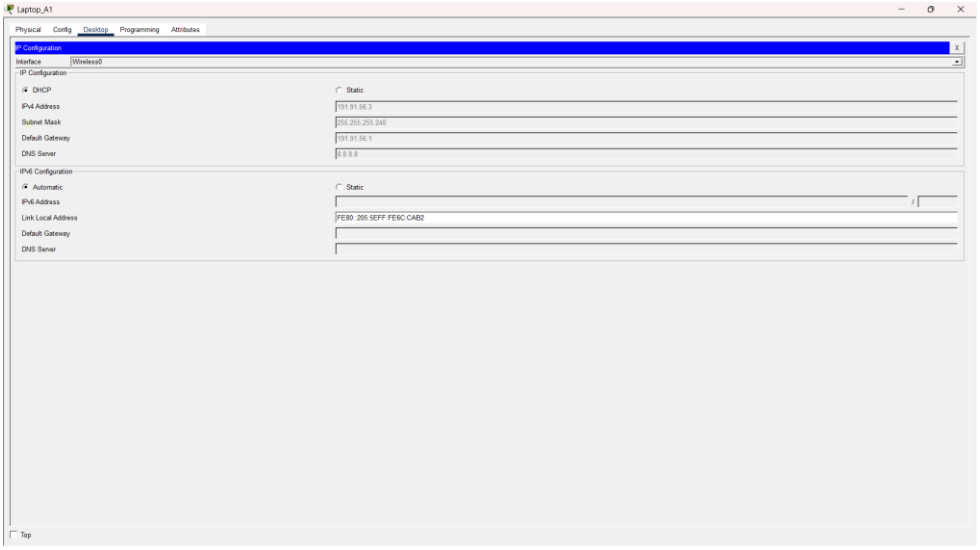

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Laptop_A1

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 191.91.56.1

Pinging 191.91.56.1 with 32 bytes of data:

Reply from 191.91.56.1: bytes=32 time=25ms TTL=255
Reply from 191.91.56.1: bytes=32 time=25ms TTL=255
Reply from 191.91.56.1: bytes=32 time=21ms TTL=255
Reply from 191.91.56.1: bytes=32 time=26ms TTL=255

Ping statistics for 191.91.56.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 21ms, Maximum = 26ms, Average = 24ms

C:\>
```

[redacted]

Laptop_A1

Physical Config Desktop Programming Attributes

Command Prompt

```
Pinging 191.91.56.18 with 32 bytes of data:

Request timed out.
Reply from 191.91.56.18: bytes=32 time=10ms TTL=127
Reply from 191.91.56.18: bytes=32 time=8ms TTL=127
Reply from 191.91.56.18: bytes=32 time=17ms TTL=127

Ping statistics for 191.91.56.18:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 8ms, Maximum = 17ms, Average = 11ms

C:\>ping 191.91.56.18

Pinging 191.91.56.18 with 32 bytes of data:

Reply from 191.91.56.18: bytes=32 time=27ms TTL=127
Reply from 191.91.56.18: bytes=32 time=25ms TTL=127
Reply from 191.91.56.18: bytes=32 time=17ms TTL=127
Reply from 191.91.56.18: bytes=32 time=12ms TTL=127

Ping statistics for 191.91.56.18:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 12ms, Maximum = 27ms, Average = 20ms

C:\>
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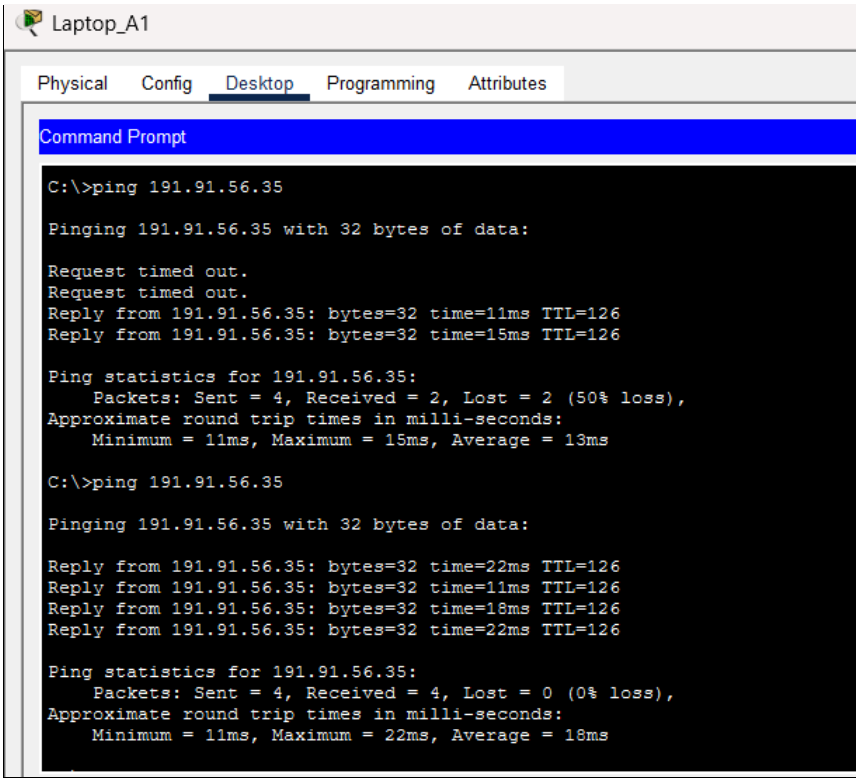

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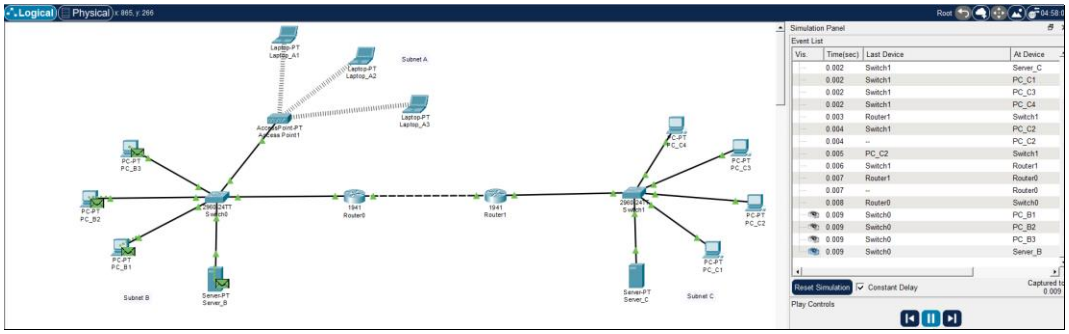

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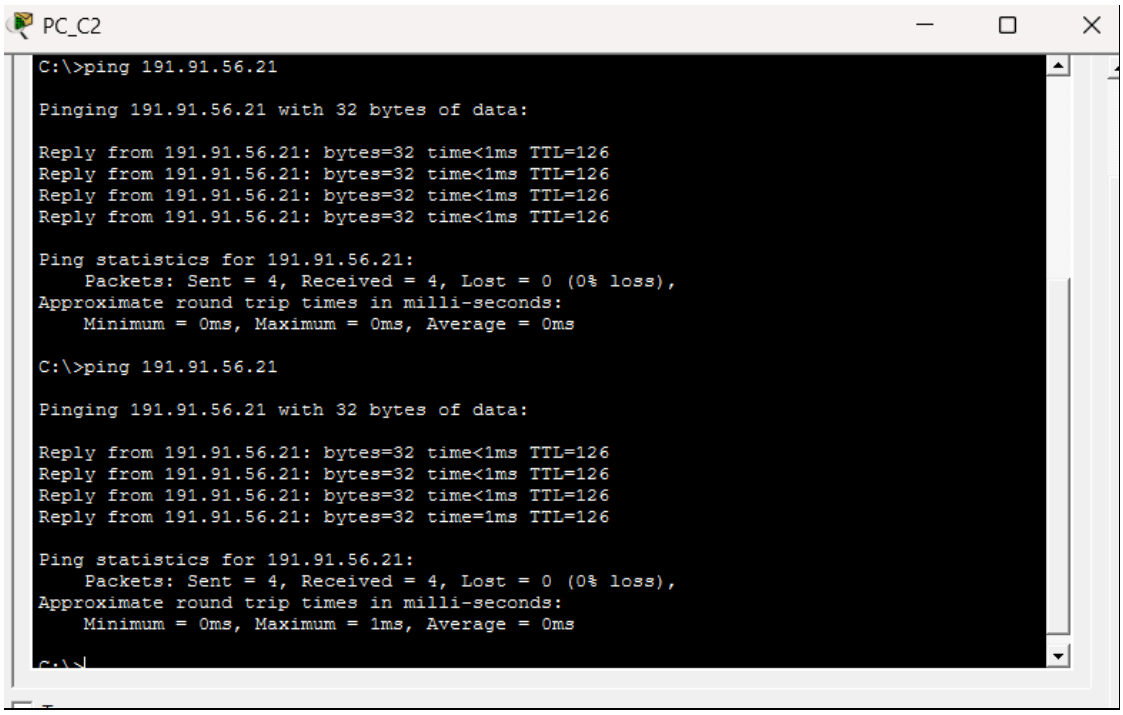

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